

PATENT RESEARCH REPORT

1. ABSTRACT

The synthesis and formulation of low-to-medium acrylonitrile carboxylated nitrile butadiene rubber (XNBR) latexes are designed to mimic the physical properties of natural rubber—specifically high elongation, low modulus, and high tensile strength—for thin-walled elastomeric articles such as medical examination gloves. The patent landscape, represented by US9926439B2, US8470422B2, and EP2010602B1, establishes a core technology based on blending two distinct XNBR terpolymers to achieve synergistic mechanical and processing performance. The primary polymerization and formulation approach involves blending a first XNBR terpolymer (which provides a lower elastic modulus and softness) with a second XNBR terpolymer (which exhibits superior film-forming capabilities) in a compositional ratio ranging from 60:40 to 40:60, with a 50:50 ratio being preferred. The resulting blended polymer composition features a low-to-medium acrylonitrile content between 17% and 45% by weight, a methacrylic acid content of less than 15% by weight, and the remaining balance being butadiene. To achieve the desired elastomeric properties, a dual-crosslinking system is employed. Ionic crosslinking of the carboxylic acid groups is controlled using 0.5 to 1.5 parts of a metallic oxide (typically zinc oxide) combined with 0 to 1.5 parts of an alkali agent providing monovalent ions. Covalent crosslinking of the butadiene segments is achieved using 0.5 to 1.5 parts sulfur and 0.5 to 1.5 parts of a rubber accelerator system, specifically exemplified by a combination of 0.5 phr diphenyl guanidine (DPG), 0.25 phr zinc mercaptobenzothiazole (ZMBT), and 0.25 phr zinc diethyldithiocarbamate (ZDEC). Process trends highlight the control of glove thickness to a thin range of 0.05 to 0.11 mm to minimize the force required for stretching. Furthermore, EP2010602B1 introduces a post-polymerization surface modification involving an exterior coating of silicone emulsion, silicone oil, or glycerin, alongside a controlled drying temperature profile of 80°C to under 100°C to optimize surface release and tactile properties.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9926439B2
- Patent Title: Nitrile rubber article having natural rubber characteristics
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018-03-27

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Elongation: 400 % (400 % of Elongation) - Dpg: 0.5 phr (0.5 phr of Dpg) - Zmbt: 0.25 phr (0.25 phr of Zmbt) - Zdec: 0.25 phr (0.25 phr of Zdec)

Relevant Experimental Evidence - Example EXAMPLE 12: Elongation: 400 %, Dpg: 0.5 phr, Zmbt: 0.25 phr, Zdec: 0.25 phr

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US8470422B2
- Patent Title: Nitrile rubber article having natural rubber characteristics
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013-06-25
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Elongation: 400 % (400 % of Elongation) - Dpg: 0.5 phr (0.5 phr of Dpg) - Zmbt: 0.25 phr (0.25 phr of Zmbt) - Zdec: 0.25 phr (0.25 phr of Zdec)

Relevant Experimental Evidence - Example EXAMPLE 12: Elongation: 400 %, Dpg: 0.5 phr, Zmbt: 0.25 phr, Zdec: 0.25 phr

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: EP2010602B1
- Patent Title: Exterior-coated nitrile rubber article having natural rubber characteristics
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016-11-02
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Temperature: 80 °C (temperature ... 80 °C) - Elongation: 400 % (400 % of Elongation) - Dpg: 0.5 phr (0.5 phr of Dpg) - Zmbt: 0.25 phr (0.25 phr of Zmbt) -

Zdec: 0.25 phr (0.25 phr of Zdec) - Modulus: 300 % (300 % of Modulus) - Strain: 300 % (300 % of Strain) - Extension: 400 % (400 % of Extension)

Relevant Experimental Evidence - Example EXAMPLE 12: Elongation: 400 %, Temperature: 80 °C, Dpg: 0.5 phr, Zmbt: 0.25 phr, Zdec: 0.25 phr, Modulus: 300 %, Strain: 300 %, Extension: 400 %

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- **Monomer Content and Ratios:** All three patents (US9926439B2, US8470422B2, and EP2010602B1) specify a carboxylated nitrile-butadiene rubber (XNBR) blend with an acrylonitrile (AN) content of 17% to 45% by weight and a methacrylic acid (MAA) content of less than 15% by weight, with butadiene making up the remainder.
- **Terpolymer Blending Strategy:** The patents consistently employ a blend of two distinct XNBR terpolymers in a compositional ratio ranging from 60:40 to 40:60 (optimally 50:50). The first terpolymer is characterized by a lower elastic modulus to impart softness, while the second terpolymer exhibits superior film-forming properties to facilitate dip-processing.
- **Crosslinking and Vulcanization Systems:** The formulation chemistry is identical across the documents, utilizing a dual-crosslinking system: ionic crosslinking via 0.5 to 1.5 parts of a metallic oxide (such as zinc oxide) combined with 0 to 1.5 parts of an alkali agent (total combined > 1.0 part), and covalent crosslinking via 0.5 to 1.5 parts sulfur and 0.5 to 1.5 parts of a multi-component accelerator system.
- **Specific Accelerator Package:** The examples in all three patents specify a highly defined accelerator package consisting of 0.5 phr diphenyl guanidine (DPG), 0.25 phr zinc mercaptobenzothiazole (ZMBT), and 0.25 phr zinc diethyldithiocarbamate (ZDEC) to achieve optimal vulcanization kinetics and physical properties.
- **Key Distinctions and Process Innovations:** While US9926439B2 and US8470422B2 focus on the core latex blend and glove article properties, EP2010602B1 introduces a critical post-processing step. It characterizes the elastomeric glove by an exterior coating of a silicone emulsion, silicone oil, glycerin solution, or fluorocarbon release agent, and specifies a drying temperature profile of 80°C to below 100°C for the gelled glove on the mold.
- **Physical Performance Targets:** All three patents target a force-strain profile mimicking natural rubber, specifically achieving a low force at 400% elongation (typically 1.5 to 1.7 N compared to 1.3 N for natural rubber) and high tensile strength (30 to 58 MPa) at reduced glove thicknesses (0.05 to 0.11 mm).

4. CONCLUSION

The synthesis of low-acrylonitrile carboxylated nitrile butadiene rubber (XNBR) with natural rubber-like characteristics is successfully realized through a synergistic blending strategy. By combining a low-modulus XNBR terpolymer with a high-film-forming XNBR terpolymer in a 50:50 ratio, manufacturers can overcome the traditional trade-off between softness and processability. Recommended synthesis and formulation parameters derived from the patent evidence include maintaining an overall acrylonitrile content of 17-45 wt%, methacrylic acid content below 15 wt%, and utilizing a dual-crosslinking system comprising 0.5-1.5 phr zinc oxide, 0.5-1.5 phr sulfur, and a balanced accelerator package of 0.5 phr DPG, 0.25 phr ZMBT, and 0.25 phr ZDEC. Additionally, incorporating an exterior silicone or glycerin coating and maintaining drying temperatures between 80°C and 100°C optimizes the surface properties and structural integrity of the resulting thin-walled (0.05-0.11 mm) elastomeric articles.

5. REFERENCES

- US9926439B2 | Nitrile rubber article having natural rubber characteristics | Not disclosed | US | 2018 | <https://patents.google.com/patent/US9926439B2/en>
- US8470422B2 | Nitrile rubber article having natural rubber characteristics | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8470422B2/en>
- EP2010602B1 | Exterior-coated nitrile rubber article having natural rubber characteristics | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP2010602B1/en>